

Probabilities and the Equation

Topics in Language and Mind: Conditionals

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1. Probability functions

When f is an n -place function and S is a set, S is closed under f iff whenever $x_1 \in S, \dots$, and $x_n \in S$, and $\langle x_1, \dots, x_n \rangle$ is in the domain of f , $f(x_1, \dots, x_n) \in S$.

A set of propositions is **Boolean-closed** (=closed under finitary Boolean operations) iff it is closed under $\neg$ (the function that maps every proposition to its negation), $\wedge$ (the function that maps any two propositions to their conjunction), and $\vee$ (the function that maps any two propositions to their disjunction).

A **probability function** is any function Pr from some Boolean-closed set of propositions D to real numbers, such that for any $A, B \in D$

- (i) $\text{Pr}(A) = 1$ if A is logically necessary
- (ii) $\text{Pr}(A) \leq \text{Pr}(B)$ if A logically entails B
- (iii) $\text{Pr}(A \vee B) = \text{Pr}(A) + \text{Pr}(B)$ if A and B are logically incompatible

Commentary:

- To make sense of the definition we need to understand some notions of “logical necessity”, “logical entailment” and “logical incompatibility” that apply to propositions. Any one of these could be defined in terms of the others in obvious ways. It’s controversial whether this should just be identified with metaphysical necessity.
- In mathematics, probability functions are standardly identified with functions from *subsets of some given set* to real numbers. If we assumed that propositions are sets of worlds, our definition would be a special case of this; but there is no need to make this assumption if we prefer not to.
- In mathematics, an additional ‘countable additivity’ condition is often built into the definition.
- Some philosophers instead use “probability function” to denote functions from *the sentences of some language* to real numbers. Given an interpretation of the language which never maps logically inequivalent sentences to the same proposition, you can go back and forth.

Some noteworthy consequences of this definition (for all A, B in the domain of Pr):

- (i) If A and B are logically equivalent, $\text{Pr}(A) = \text{Pr}(B)$
- (ii) $\text{Pr}(A) \leq 1$
- (iii) $\text{Pr}(A) \geq 0$

- (iv) $\Pr(A) + \Pr(\neg A) = 1$
- (v) $\Pr(A \wedge B) + \Pr(\neg A \wedge B) = \Pr(B)$
- (vi) $\Pr(A \vee B) = \Pr(A) + \Pr(B) - \Pr(A \wedge B)$
- (vii) If $\Pr(A \wedge B) = 0$, $\Pr(A) \leq \Pr(\neg B)$
- (viii) If $\Pr(A \vee B) = 1$, $\Pr(\neg A) \leq \Pr(B)$

When $\Pr$ is a probability function, $\Pr(A | B)$ is defined to be $\Pr(A \wedge B) / \Pr(B)$ whenever $\Pr(B) > 0$.

Exercise: show that $\Pr(A) = \Pr(A \wedge B | B)\Pr(B) + \Pr(A \wedge \neg B | \neg B)\Pr(\neg B)$ whenever $0 < \Pr(B) < 1$

When $\Pr$ is any probability function with domain D and B is any proposition for which $\Pr(B) > 0$, $\Pr_B$ is defined to be the function on D such that for any $A \in D$, $\Pr_B(A) = \Pr(A | B)$.

Exercise: show that $\Pr_B$ is a probability function.

2. Objective chances

Let the objective chance function at t , Ch_t , be the function that maps every proposition to its objective chance at t .

Claim: for every t , Ch^t is a probability function.

Facts entirely about (the intrinsic character of) history before t have chance 1 at t .

How do chances at different times relate to one another? At least in the special case where the complete truth H about (the intrinsic character of) history between t and $t+$ has chance > 0 at t , we have: $\text{Ch}^{t+} = \text{Ch}^t_H$. ('Chances evolve by conditionalization').

3. Credences and probabilism

Credence = degrees of belief = degrees of confidence, measured on a scale from 0 to 1.

- The credence function of S at t = the function that maps every proposition (in which S has a credence at t) to S 's credence in it in t .

What is it to have a certain credence in a proposition? A hard question for philosophy of mind. But it is worth mentioning an influential "behaviourist" tradition which identifies having a credence with having a certain behavioral dispositions. Cartoon version: to have credence x in P is to be willing to pay anything up to $\$x$ for a bet that pays \$1 if P and \$0 otherwise.

Do ordinary people actually have credences in most of the propositions they think about? The fact that credences are real numbers makes some philosophers think not. But maybe if you are the kind of philosopher who is comfortable with 'There is a least number of hairs consistent with not being bald', you should be similarly comfortable

with 'There is a real number that is Joe Schmoe's current credence that it will rain tomorrow'.

Probabilism: at any time, the credence function of an (ideally) rational person is a probability function.

- Setting aside objections to the claim that rational people should have credence functions at all, the most prominent challenges to probabilism target its implication that if one is ideally rational one assigns credence 1 to all logically necessary propositions.
- However those who dispute this are often happy to treat it as a useful idealization.

(Incremental) Bayesianism: if S has credence function Cr at t and is (ideally) rational throughout the interval from t to $t+$, then for some proposition E , S 's credence function at $t+$ is Cr_E . ('Credences evolve by conditionalization').

- What proposition should one conditionalize on? Bayesians say: "on one's total evidence". But 'evidence' here is used in a somewhat technical sense.
- Probabilists who reject Incremental Bayesianism want to allow for (i) rational forgetting, and/or (ii) evolution of credence without becoming absolutely certain of anything. 'Bayesian' is often used loosely in a way that covers some alternatives to Incremental Bayesianism.